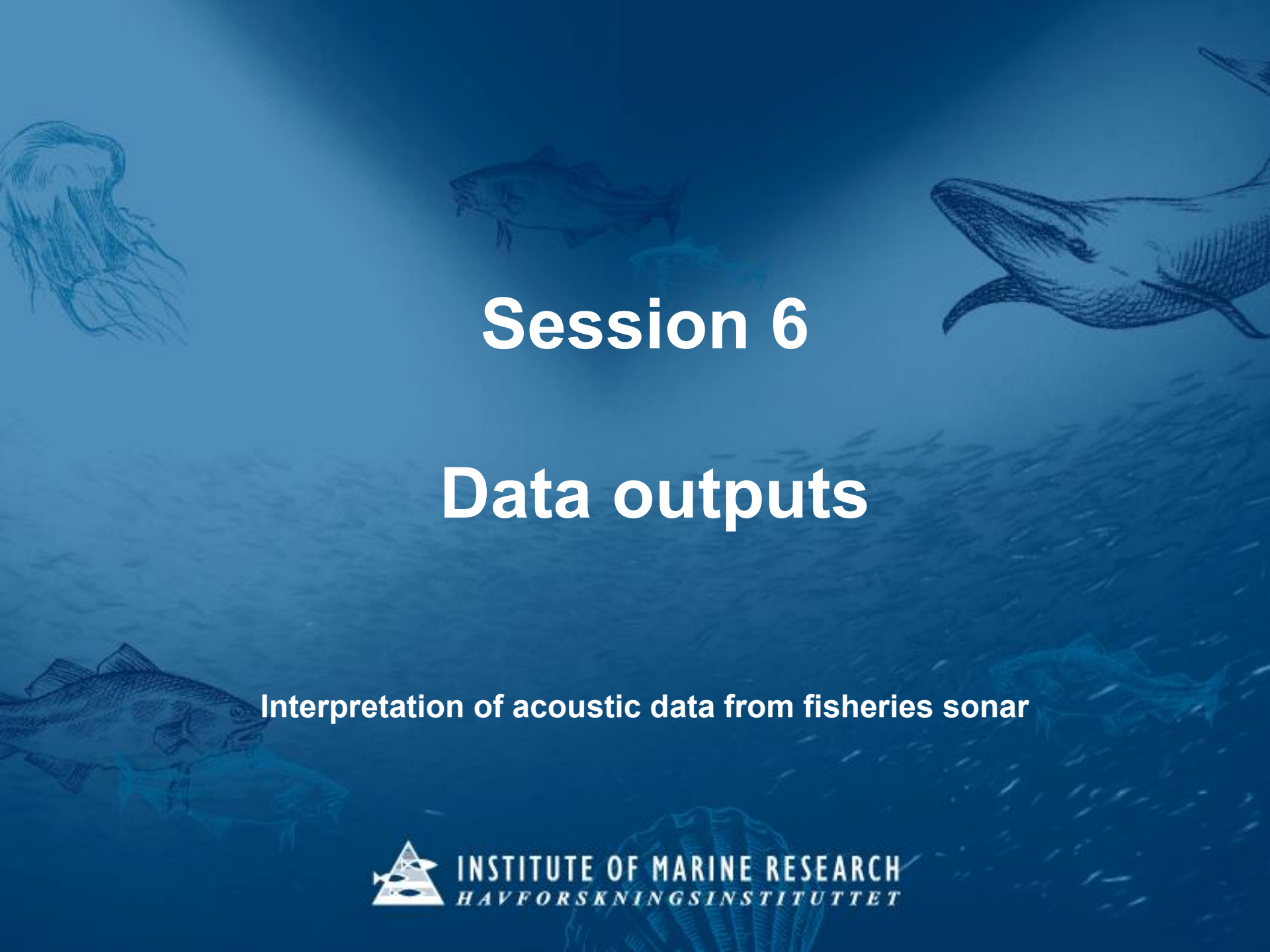




Session 6

Data outputs

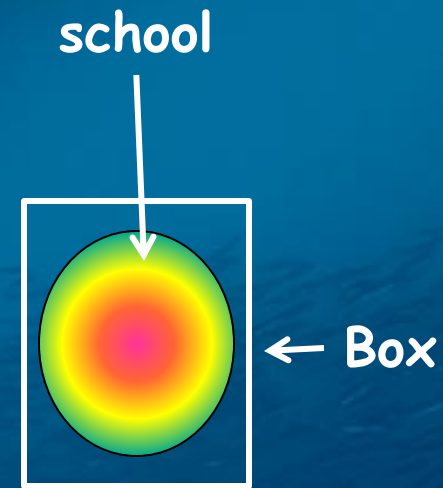
Interpretation of acoustic data from fisheries sonar



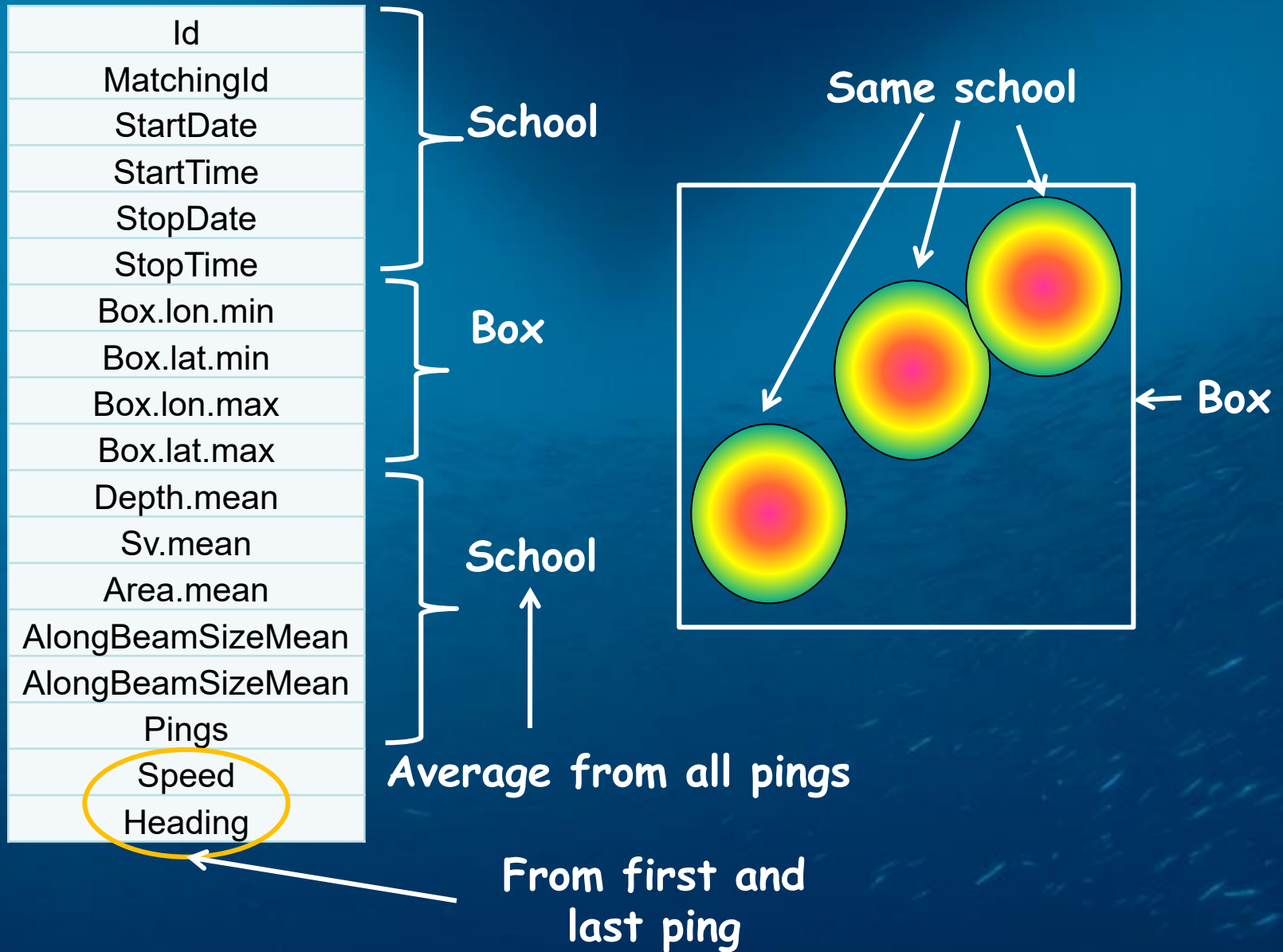
INSTITUTE OF MARINE RESEARCH
HAVFORSKNINGSINSTITUTTET

Description of output data per ping

Id	School
Date	
Time	
File	
Center.lon	
Center.lat	Box
Center.dep	
Box.lon.min	
Box.lat.min	
Box.lon.max	
Box.lat.max	School
Sv.mean	
Area	
Ship.lon	Ship
Ship.lat	
Ship.speed	
Ship.heading	
Trans.tilt	School
AlongBeamSize	
AlongRingSize	



Description of output data per school



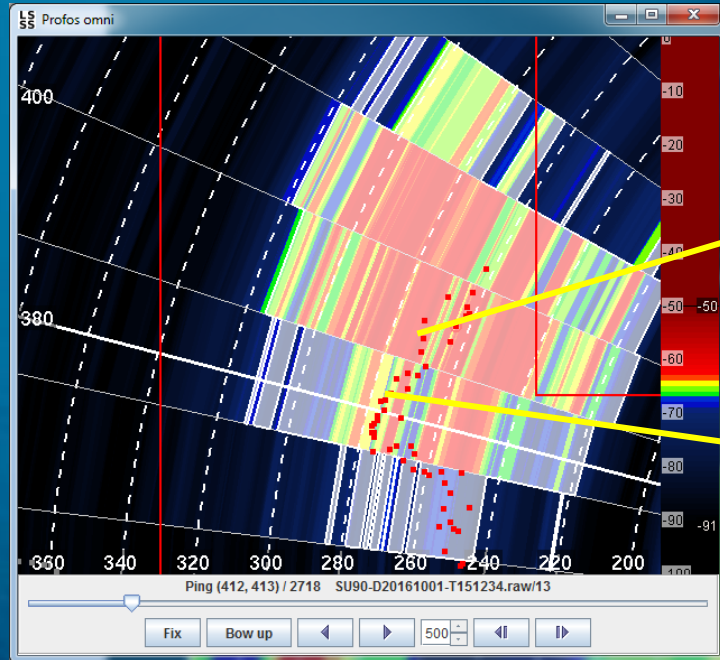
Description of output data per ping for echointegration

Id	Date	Time	File	Center.lon	Center.lat	Center.dep	Box.lon.min	Box.lat.min	Box.lon.max	Box.lat.max	Sv.mean	Area	Ship.lon	Ship.lat	Ship.speed	Ship.heading	Trans.tilt	AlongBeamSize	AlongRingSize
1	11.07.2019	15:29:54.54	SU90-D20190711-T152954.raw/0	-0.39472	56.61087	24.93	-0.39487	56.61069	-0.39455	56.61101	-67.62	135.2	-0.39361	56.61053	0	169.8	-13.01	4.84	39.82
1	11.07.2019	15:29:56.15	SU90-D20190711-T152954.raw/2	-0.39467	56.6108	23.81	-0.39481	56.61071	-0.39456	56.6109	-70.57	93.2	-0.39361	56.61051	1.38	169.3	-13.01	7.45	22.35
1	11.07.2019	15:29:56.97	SU90-D20190711-T152954.raw/3	-0.39464	56.61086	24.18	-0.3948	56.61071	-0.39436	56.61105	-63.22	135.7	-0.39361	56.61051	1.5	168.6	-13.01	6.15	46.13
1	11.07.2019	15:29:57.76	SU90-D20190711-T152954.raw/4	-0.39466	56.61085	24.56	-0.39489	56.6107	-0.3943	56.61108	-67.78	166.1	-0.3936	56.61049	2.88	167.9	-13.01	10.62	47.45
1	11.07.2019	15:29:58.58	SU90-D20190711-T152954.raw/5	-0.39469	56.61083	25.03	-0.39487	56.61073	-0.39449	56.61096	-63.72	136	-0.3936	56.61047	2.74	167.2	-13.01	10.8	31.19
1	11.07.2019	15:29:59.39	SU90-D20190711-T152954.raw/6	-0.39467	56.61083	25.13	-0.39488	56.61072	-0.3945	56.61095	-66.51	132.1	-0.3936	56.61045	2.63	166.5	-13.01	7.64	24.27
1	11.07.2019	15:30:00.28	SU90-D20190711-T152954.raw/7	-0.39473	56.61081	25.87	-0.3949	56.61069	-0.39453	56.61093	-68.14	160	-0.39359	56.61043	2.63	165.8	-13.01	9.31	24.77
1	11.07.2019	15:30:01.09	SU90-D20190711-T152954.raw/8	-0.39472	56.61084	26.41	-0.3949	56.61073	-0.39448	56.61096	-74.79	187.9	-0.39359	56.61041	1.3	165.1	-13.01	11.73	33.32
1	11.07.2019	15:30:02.00	SU90-D20190711-T152954.raw/9	-0.3947	56.61085	26.34	-0.39489	56.61073	-0.39458	56.61092	-63.05	114.8	-0.39359	56.61041	1.34	164.3	-13.01	7.08	25.7
N/A	11.07.2019	15:29:55.36	SU90-D20190711-T152954.raw/1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.39361	56.61053	1.38	169.8	-13.01	N/A	N/A
N/A	11.07.2019	15:30:07.34	SU90-D20190711-T152954.raw/15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.39356	56.61029	3.01	160.2	-13.01	N/A	N/A
N/A	11.07.2019	15:30:08.14	SU90-D20190711-T152954.raw/16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.39355	56.61026	1.53	159.6	-13.01	N/A	N/A
N/A	11.07.2019	15:30:08.95	SU90-D20190711-T152954.raw/17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.39355	56.61026	1.4	159	-13.01	N/A	N/A
N/A	11.07.2019	15:32:14.76	SU90-D20190711-T153155.raw/23	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.39023	56.60832	2.67	121.2	-13.01	N/A	N/A
N/A	11.07.2019	15:33:21.36	SU90-D20190711-T153316.raw/6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.38757	56.60779	3.34	104.2	-13.01	N/A	N/A
N/A	11.07.2019	15:34:57.31	SU90-D20190711-T153457.raw/0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.38365	56.60693	2.65	117.2	-13.01	N/A	N/A
N/A	11.07.2019	15:34:58.22	SU90-D20190711-T153457.raw/1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.38365	56.60693	1.5	117.2	-13.01	N/A	N/A
N/A	11.07.2019	15:34:59.04	SU90-D20190711-T153457.raw/2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-0.38362	56.60691	1.51	116.6	-13.01	N/A	N/A

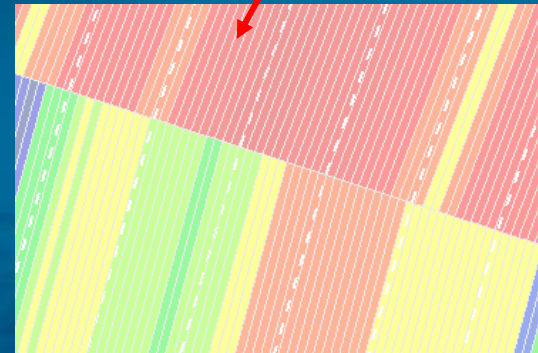
Pings with no schools, needed for echointegration computations



Sample data from each voxel (3D pixel)



sample data



Data from each sample by
beam and sampling range



Example of sample data export

```
# Lines starting with # are comments
# The format of and definitions in this file may change in future versions
# of LSSS, last changed in LSSS version 1.7.1
# Export time: Mon Dec 12 11:59:31 CET 2016, LSSS version 1.11.0
#
# School id: 3
# Ping: SU90-D20161001-T151234.raw/13
# Beams are numbered starting at 0
# The sample values are Sv
#
# Range[m],Beam21[dB re m2],Beam22[dB re m2],Beam23[dB re
# m2],Beam24[dB re m2],Beam25[dB re m2],Beam26[dB re m2]
Range,Beam21,Beam22,Beam23,Beam24,Beam25,Beam26
213.26,,,,,-69.45,,
213.44,,,,,-70.03,,
213.63,,,,,-70.93,,
213.82,,,,,-72.14,,
214.00,,,,,-73.63,,
214.19,,,,,-75.34,,
214.37,,,,,-77.17,,
214.56,,,,,-78.60,,
214.75,,,,,-78.59,,
214.93,,,,,-77.30,-76.95,,
215.12,,,,,-75.29,-74.67,,
215.31,,,,,-73.54,-72.38,,
215.49,,,,,-71.99,-70.28,,
215.68,,,,,-70.69,-68.46,,
215.86,,,,,-69.66,-66.99,,
216.05,,,,,-68.92,-65.88,,
216.24,,,,,-68.40,-65.10,,
216.42,,,,,-68.06,-64.60,,
```

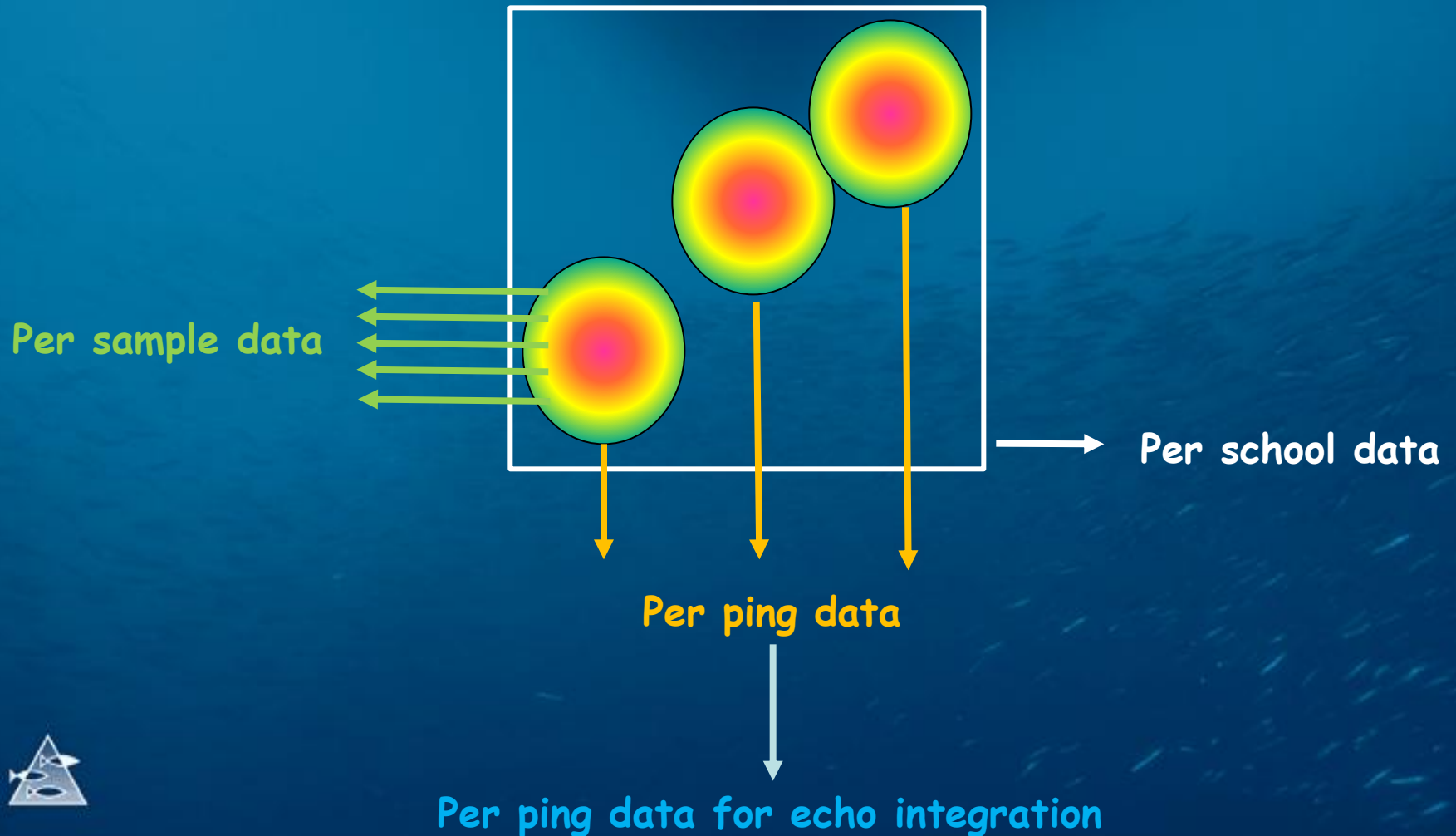
ca. 20 cm

Range	Beam21	Beam22	Beam23	Beam24	Beam25	Beam26
221.08	-75.68	-66.52	-67.94	-74.71		
221.27	-75.69	-76.26	-66.88	-66.85	-72.8	
221.45	-74.55	-75.83	-67.49	-65.91	-71	
221.64	-72.44	-73.96	-67.77	-65.19	-69.72	-72.73
221.82	-70.25	-71.49	-67.08	-64.78	-68.92	-74.6



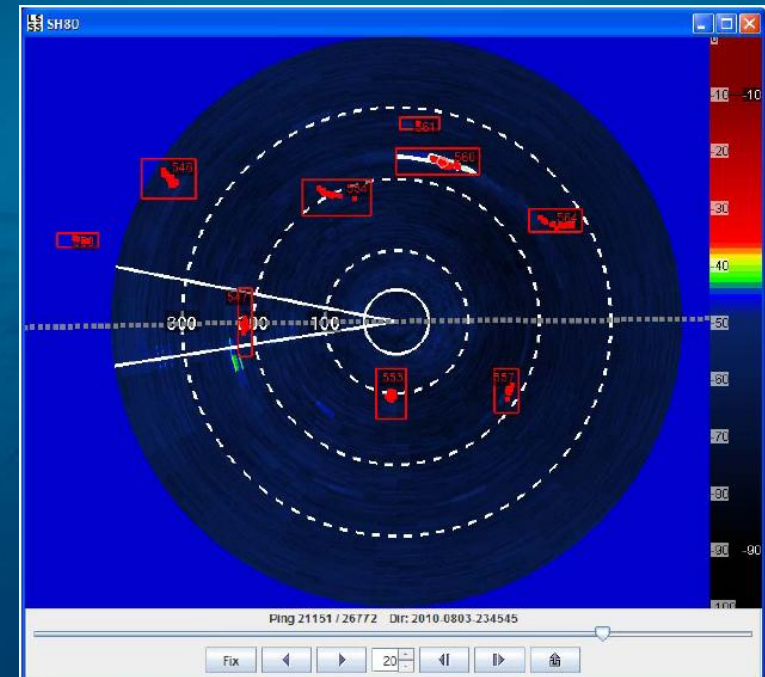
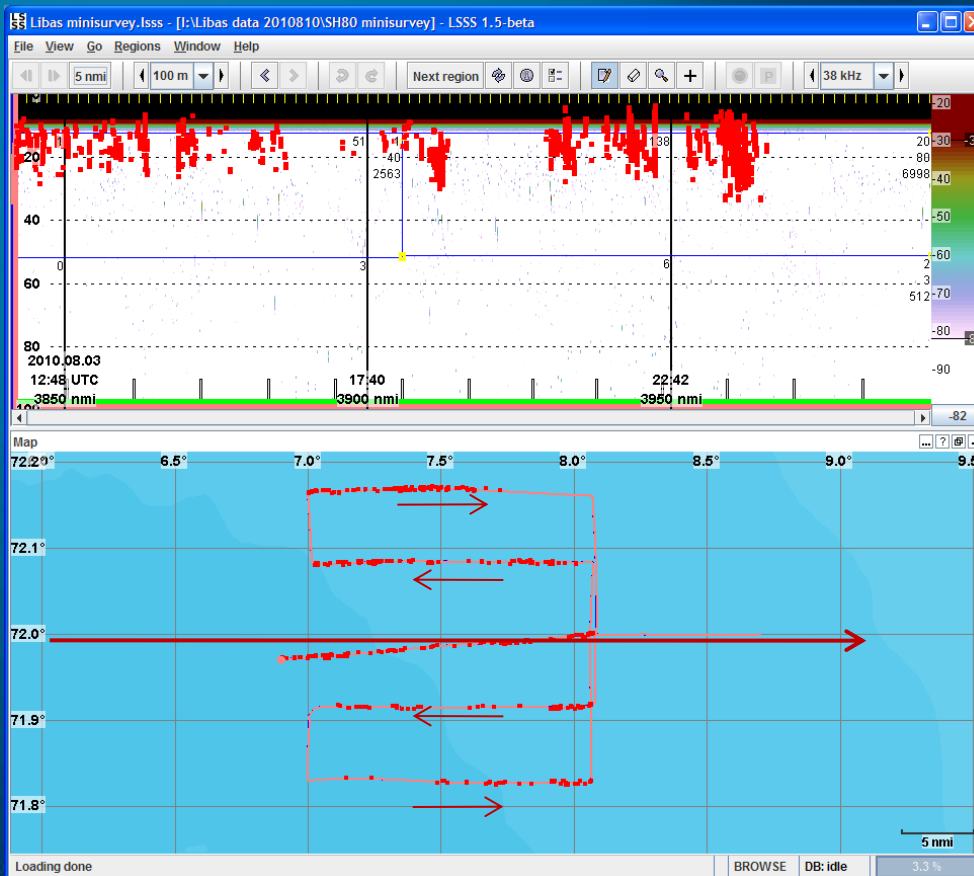
Summary of outputs

One school detected in three pings



Examples of scutinized data

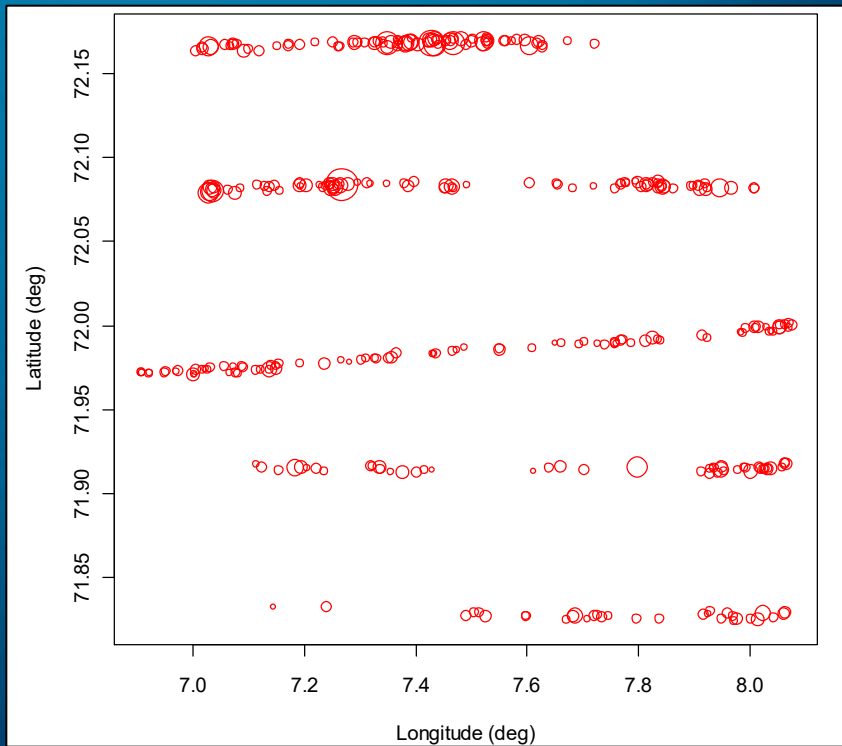
Mackerel schools in minisurvey (5 transects * 20 nmi)



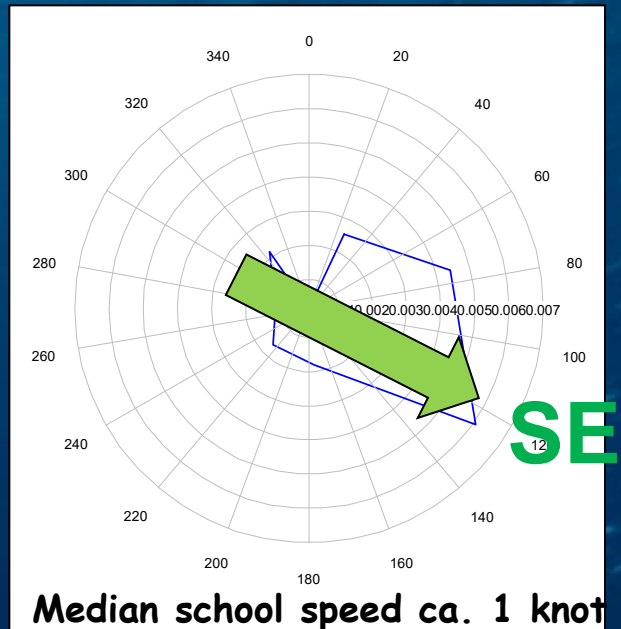
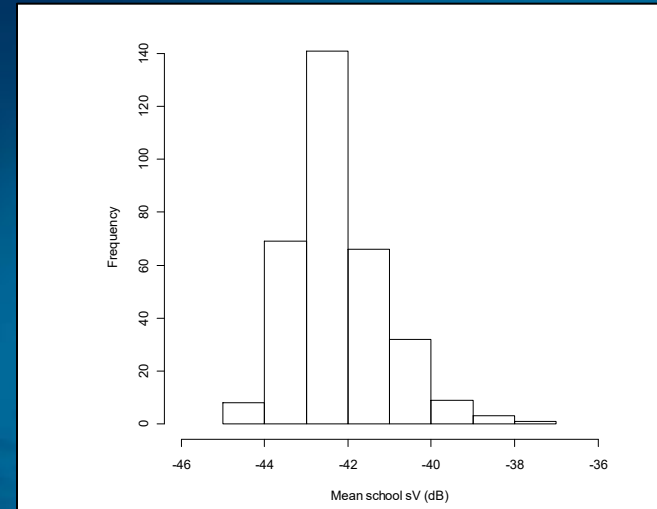
Example of mackerel schools

Examples of scutinized data

Volume backscattering coefficient (s_v)



Mean school s_v

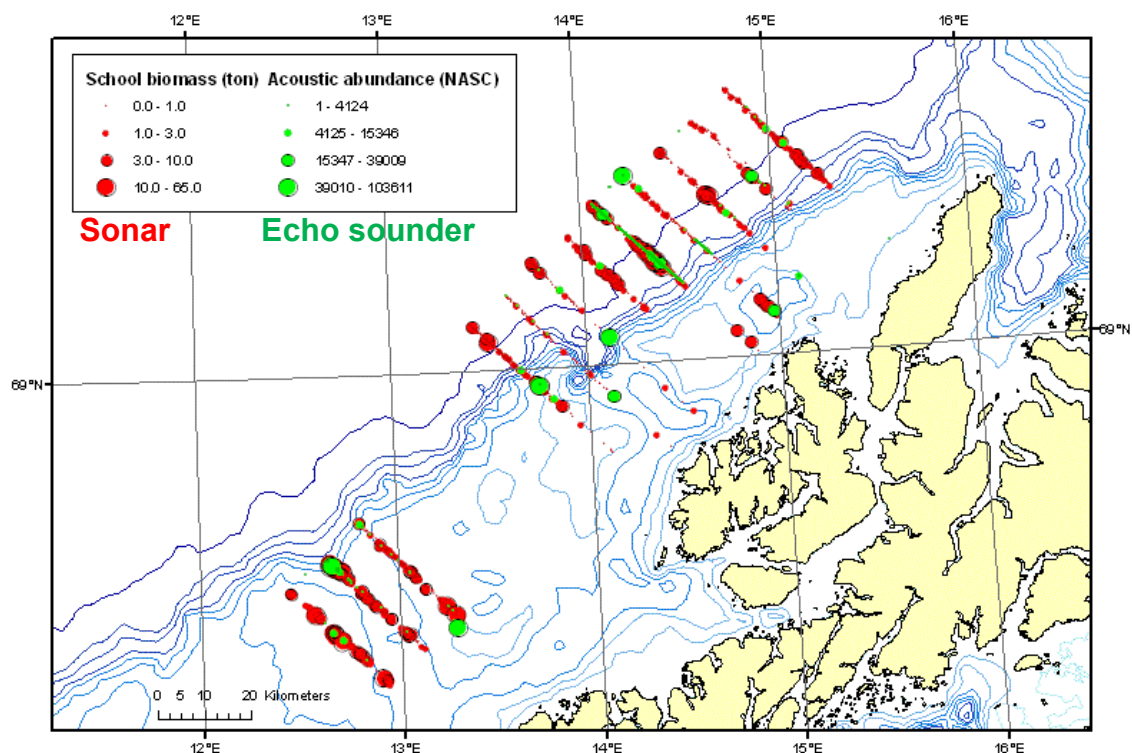


School direction

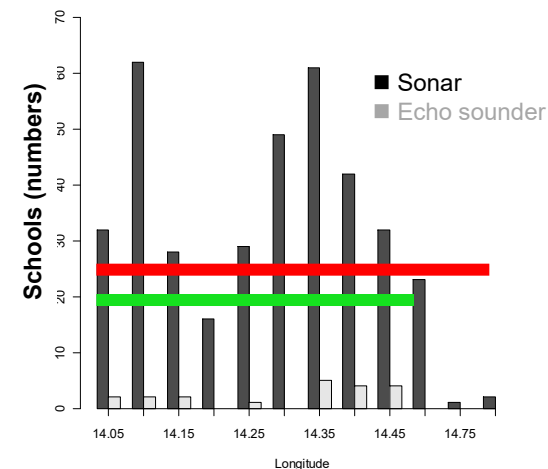
Median school speed ca. 1 knot



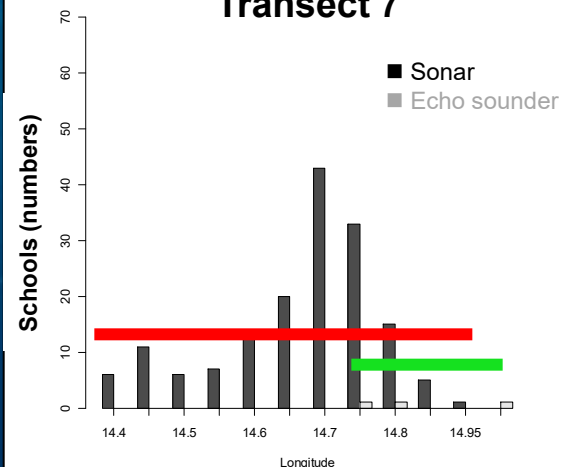
Herring schools detected by sonar and echo sounder in Vesterålen, July 2009 (Effect of seismic investigations project)



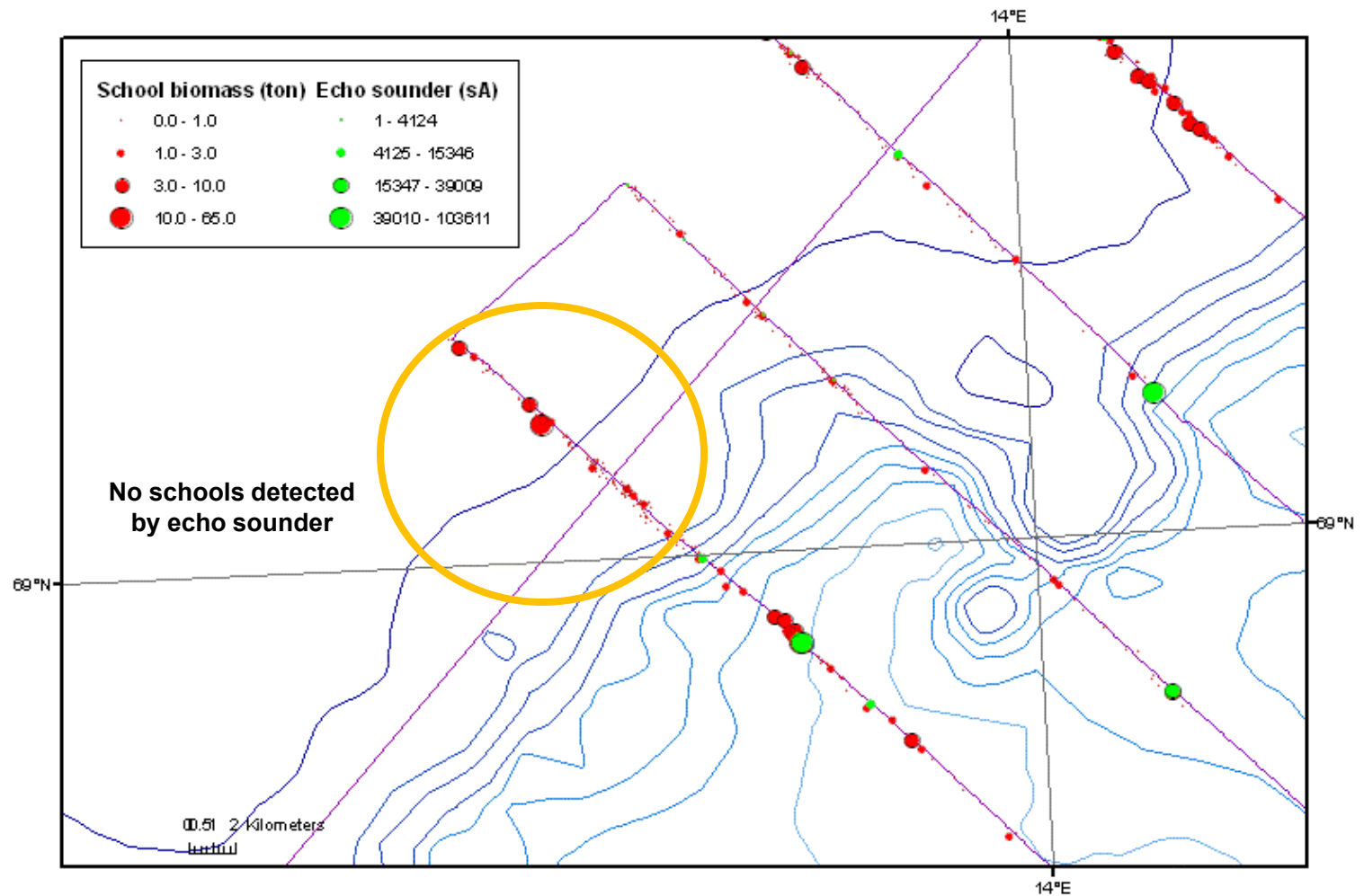
Transect 5



Transect 7

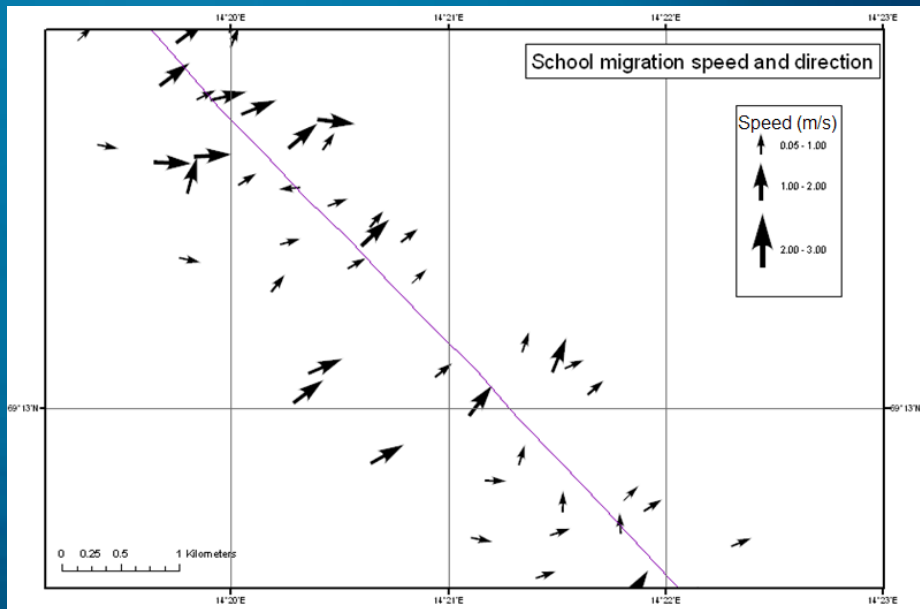


Detail of Transect 1 in Coverage 1 (22-23/06/2009)

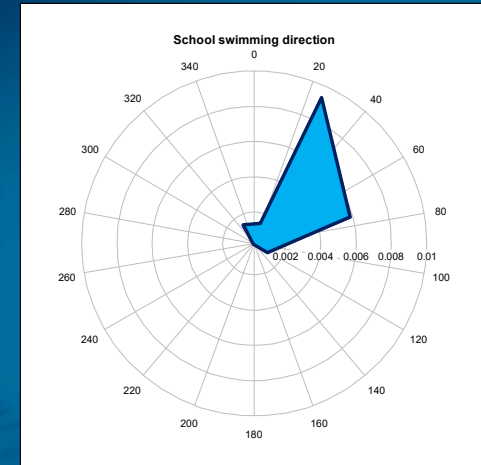


Herring school migration in Vesterålen, July 2009

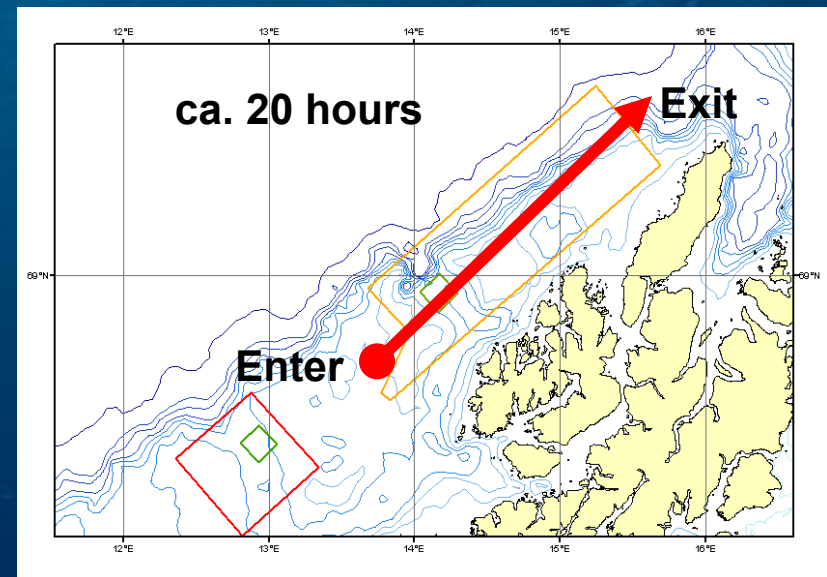
Schools swimming speed and direction along a transect



Mean swimming direction



Herring school swimming migration in in study area



Statistics of herrings schools observed by sonar in three coverages

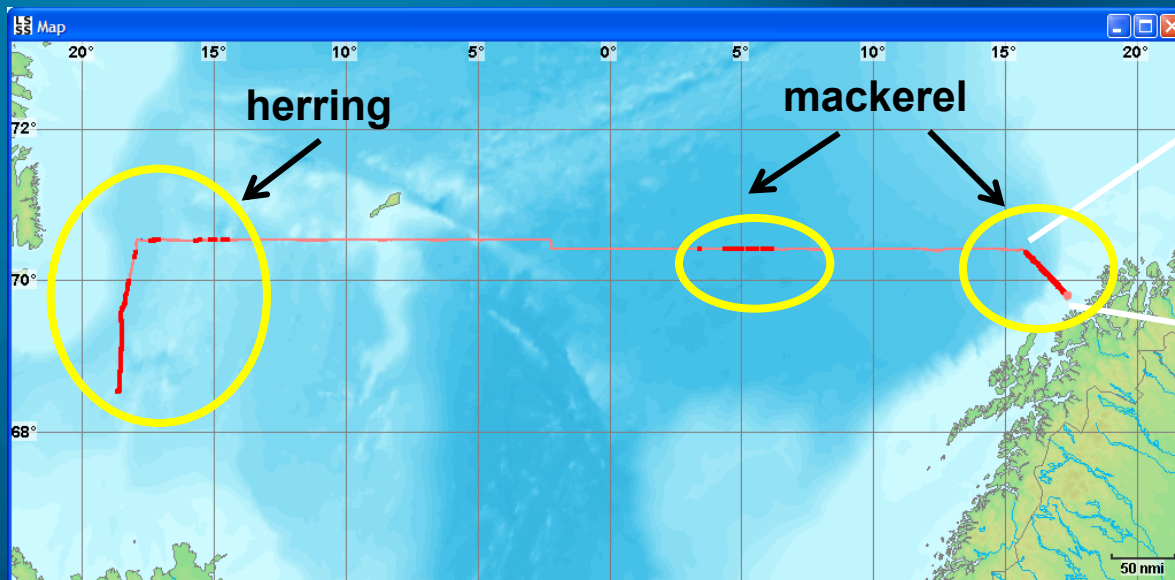
Seismic shootings



	Coverage 1	Coverage 3	Coverage 5
Total school number	934	757	45
Total biomass (ton)	24000	37000	500
School density (schools/nmi ²)	23	19	1
Mean school biomass (ton)	1.1(2.2)	2.1(3.9)	0.5(0.4)
Mean school area (m ²)	326(363)	556(514)	218(110)
Median school direction (deg)	47.9	57.7	54.7
Mean school speed (knots)	1.1(0.5)	1.5(0.5)	1.0(0.5)

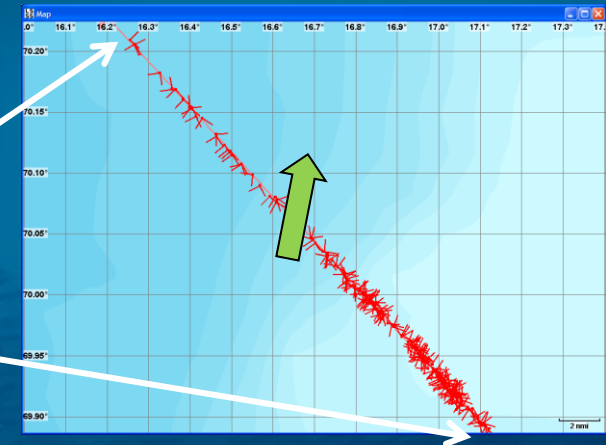


Scrutinized sonar data in one transect during mackerel ecosystem survey (> 2000 nmi)

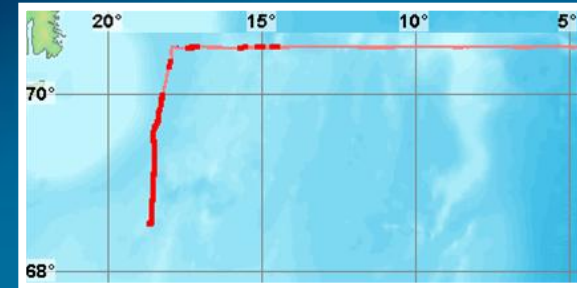
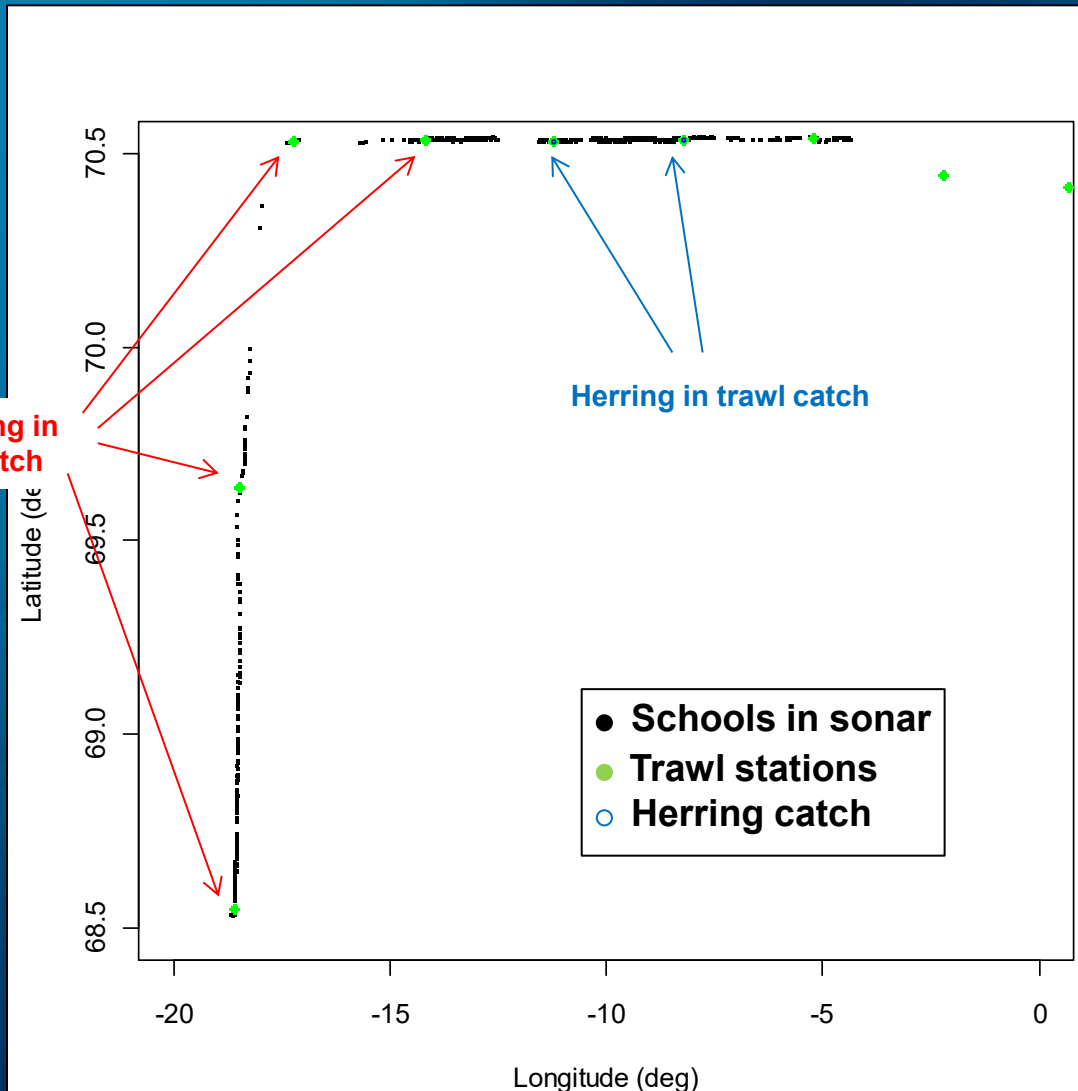


ca. 1800 schools scrutinized

School direction



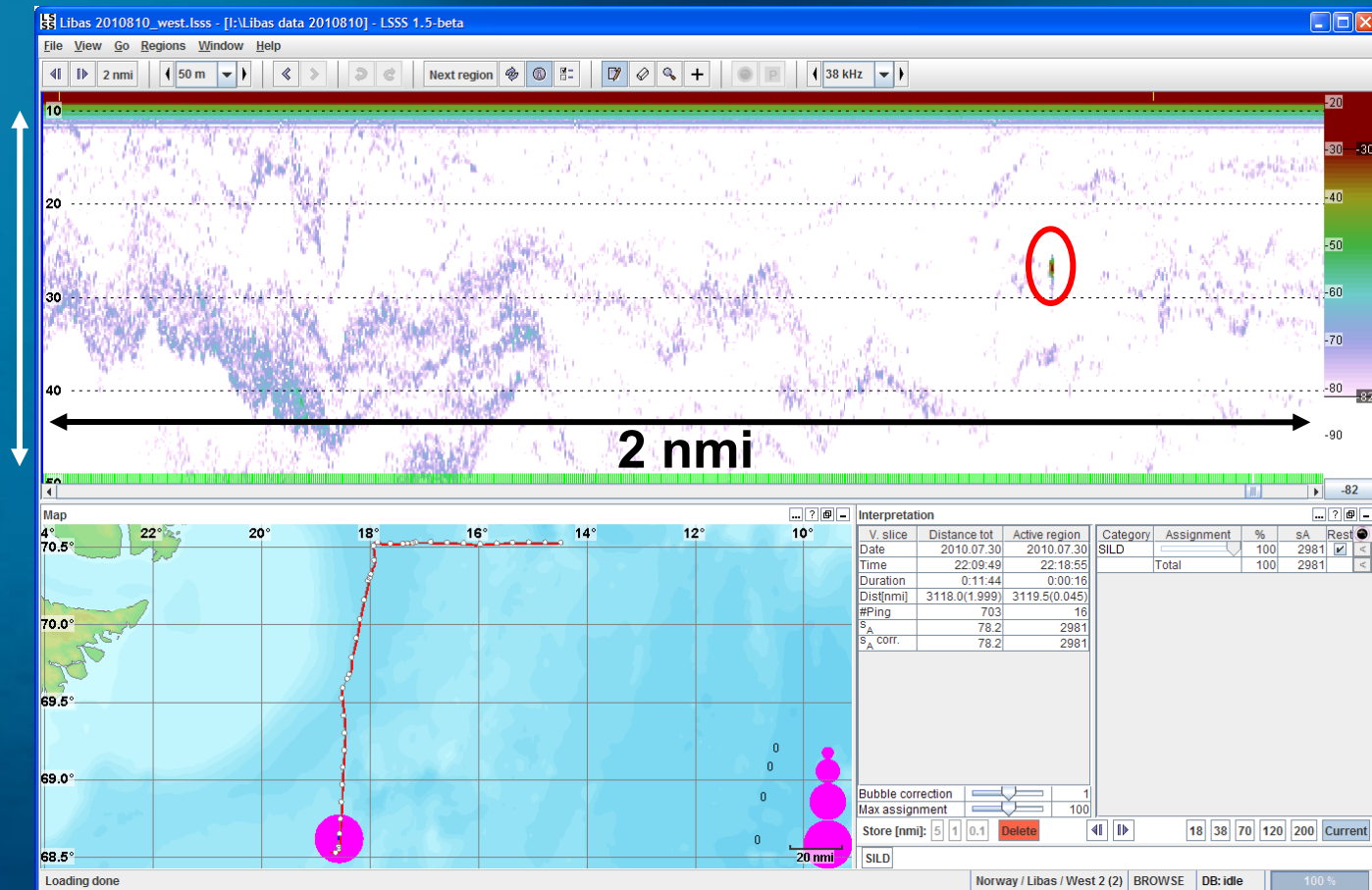
Herring schools detected by the sonar but not catch by the trawl or scrutinized in the echo sounder data, Norwegian sea summer 2010



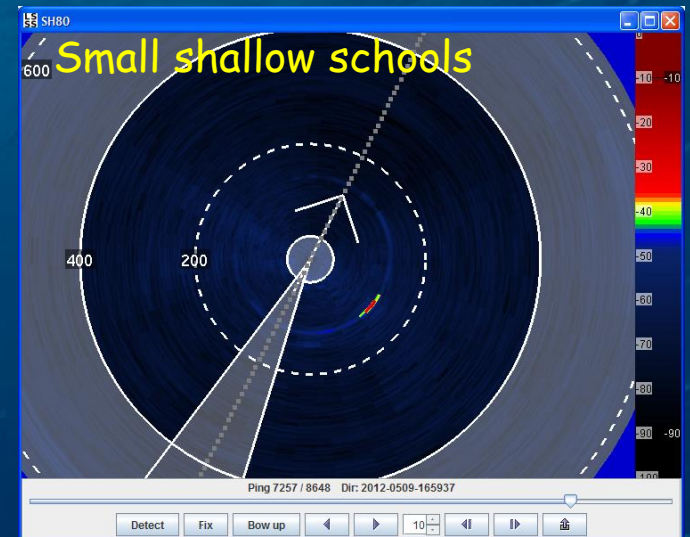
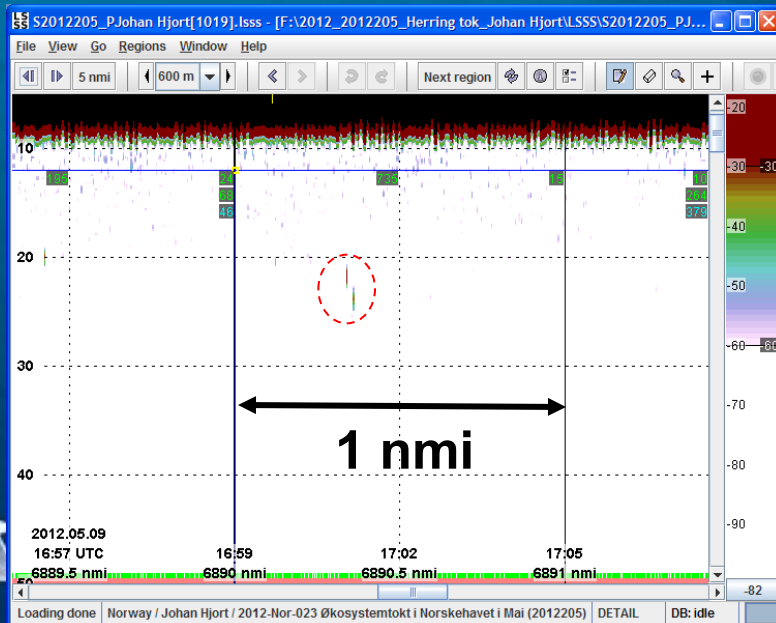
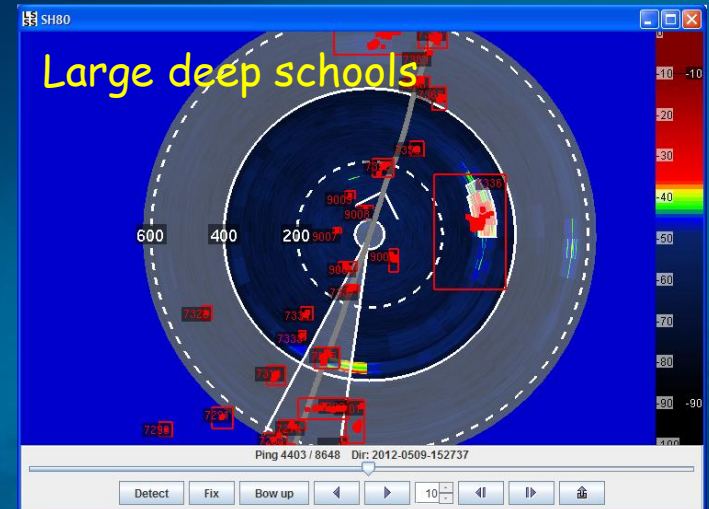
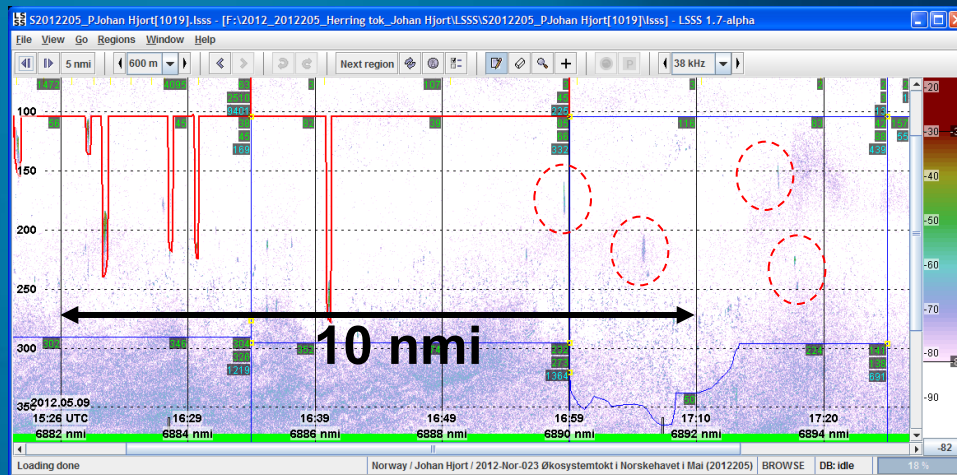
No herring allocated in echo sounder data



Detail of echogram showing a small herring school

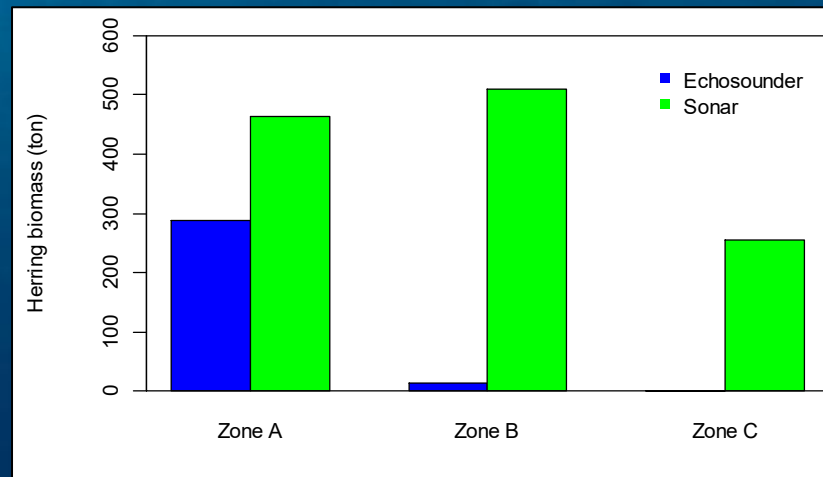
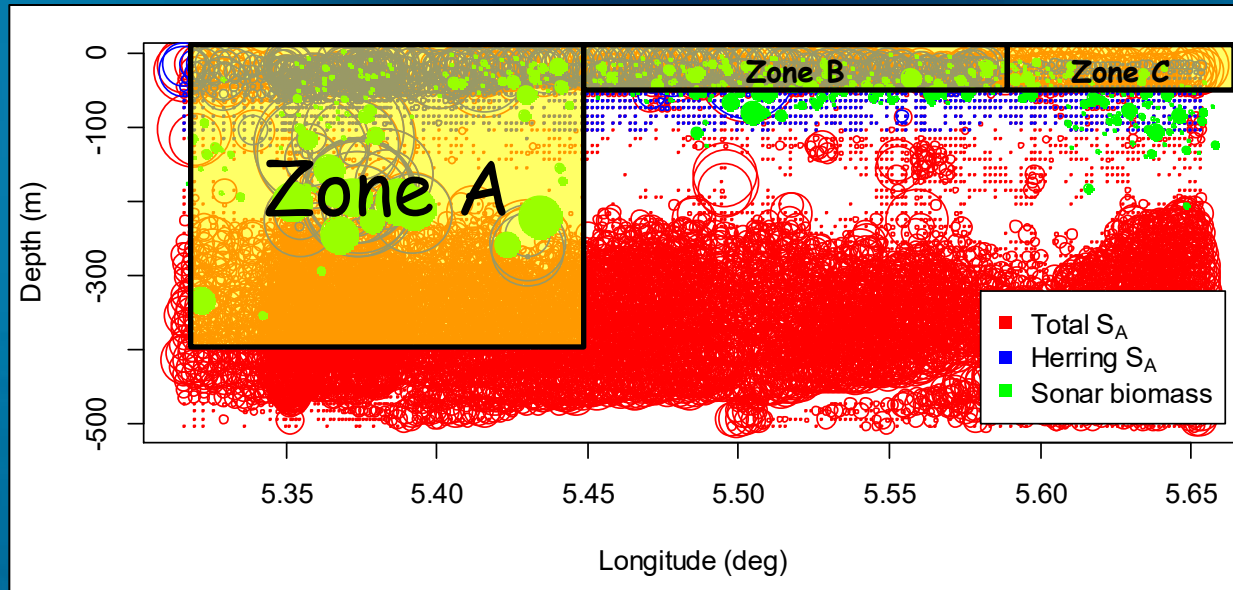


Herring schools detected by sonar and echo sounder in Norwegian Sea, May 2012



Herring schools detected by sonar and echo sounder in Norwegian Sea, May 2012

(International ecosystem survey in the Nordic Seas)



- Calibrated echo sounder
- Uncalibrated sonar

